

ADITYA MULAY

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B.Tech Computer Science (Cloud Computing & AI) | MIT ADT University, Pune
Passing Year: 2026

Internship Experience – CerebroSpark Innovations (Jan 2026 to April 2026)

Project	Problem Statement	Technologies	Output / Use
Vision-Based Object Following System	Drones lacked real-time autonomous target tracking without GPS.	Python, OpenCV, CSRT Tracker, MoSSE Tracker, MAVLink, PX4/ArduPilot	Drone that autonomously follows a moving object using live camera feed.
Straight-Line Autonomous Flight System	Manual flight control was error-prone for precise linear paths.	Python, MAVLink, PX4/ArduPilot, Flight Controllers	Automated flight system enabling drones to fly accurate straight-line trajectories.
ADS-B Airspace Awareness System	Small drones had no way to detect nearby aircraft or share position securely.	Raspberry Pi 4, RTL-SDR, ADS-B, MAVLink, LoRa, AES-256-GCM	Real-time airspace monitoring with encrypted inter-drone communication.
AprilTag Precision Landing System	Drone landing accuracy was poor without visual marker guidance.	Python, OpenCV, AprilTag, MAVLink, PX4/ArduPilot, PID Controllers	Centimeter-accurate autonomous landing using visual fiducial markers.

Internship Experience – Accenture (May 2025 to July 2025)

Project	Problem Statement	Technologies	Output / Use
Enterprise Web App Modernization	Legacy enterprise UI was slow, hard to maintain, and non-responsive.	Java, Spring Boot, Maven, Thymeleaf, Bootstrap, REST APIs	Modernized, responsive web application with improved performance and maintainability.
Business Process Automation (RPA)	Repetitive manual workflows were slowing business operations.	Blue Prism, Docker, Agile/Scrum	Automated end-to-end business processes, reducing manual effort and errors.

Academic Projects

Project	Problem Statement	Technologies	Output / Use
Aurora – AI Mental Health Platform	Lack of accessible, AI-driven early mental health support for users.	Python, NLP, Generative AI APIs	Funded AI platform providing personalized mental health conversations and insights.
RetinologyAI – Diabetic Retinopathy Detection	Manual retinal scan diagnosis is slow and expert-dependent.	Python, PyTorch, ResNet50, Tkinter	Deep learning model detecting diabetic retinopathy from retinal images with high accuracy.
Aneka Distributed Computing Research	Need to benchmark distributed task scheduling at the university level.	Aneka Platform, Distributed Computing	Research lab established at MIT ADT; featured in NAAC institutional review.
Audoria	No intelligent tool for analyzing and recommending music based on audio features.	Python, AI/ML, Music Intelligence	AI-powered music intelligence platform for analysis and recommendation.
Almost Died (Android Game)	Exploring game development and physics-based interaction on mobile.	Godot Engine, Android Development	Published Android game with physics-based gameplay mechanics.

Location Readiness

Preferably: Pune. Ready to relocate.
Open to Work from Office, Hybrid, and Remote opportunities.